Eigenvalues & Eigenvectors

Worked Examples Workbook

20 handpicked, fully solved problems · Moderate difficulty

Topics: Eigenvalue computation · Eigenvector finding · Linear independence · Arithmetic & Geometric multiplicity · Real symmetric matrices · Diagonalizability · Rank & characteristic polynomial reasoning

All solutions follow methods and reasoning from the Linear Algebra lecture series. Each problem is worked step-by-step with complete justification.

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Section 1 Eigenvalue Computation

In these problems you practice forming the characteristic equation det(A - lambda*I) = 0 and solving the resulting polynomial. Remember: for triangular matrices the eigenvalues are just the diagonal entries.

Question 1

Topic: Eigenvalue Computation — 2x2 | Level: Moderate

Find all eigenvalues of A = [[4, 1], [2, 3]].

SOLUTION

Step 1 – Form det(A - lambda*I) = 0

```
det([[4-L, 1], [2, 3-L]]) = 0
(4-L)(3-L) - (1)(2) = 0
12 - 4L - 3L + L^2 - 2 = 0
L^2 - 7L + 10 = 0
```

Expand the 2x2 determinant and collect powers of lambda (written L).

Step 2 – Factor the characteristic polynomial

```
L^2 - 7L + 10 = (L - 5)(L - 2) = 0
```

We need two numbers that multiply to 10 and add to -7: they are -5 and -2.

Final Answer: lambda_1 = 5, lambda_2 = 2

Key Insight: Two distinct eigenvalues guaranteed means two linearly independent eigenvectors exist (Theorem 1).

Question 2

Topic: Eigenvalue Computation — 3x3 Upper Triangular | Level: Moderate

Find all eigenvalues of A = [[2, 3, -1], [0, 5, 4], [0, 0, 2]].

SOLUTION

Step 1 – Recognize triangular structure

A is upper triangular. For any triangular matrix the eigenvalues are exactly the diagonal entries — no determinant expansion needed.

Step 2 – Read eigenvalues from the diagonal

```
Diagonal entries: 2, 5, 2
Characteristic polynomial: (2-L)(5-L)(2-L) = (2-L)^2 * (5-L) = 0
```

Lambda = 2 appears twice (arithmetic multiplicity 2); lambda = 5 appears once (AM = 1).

Final Answer: lambda_1 = 2 (AM = 2), lambda_2 = 5 (AM = 1)

Key Insight: Triangular matrix shortcut: eigenvalues = diagonal entries. Always verify by checking the shape.

Question 3

Topic: Eigenvalue Computation — Characteristic Polynomial Factoring | Level: Moderate

The characteristic polynomial of a 4x4 matrix A is:

p(lambda) = lambda^4 - 6*lambda^3 + 11*lambda^2 - 6*lambda

Find all eigenvalues and their arithmetic multiplicities.

SOLUTION

Step 1 – Factor out common lambda

```
p(L) = L(L^3 - 6L^2 + 11L - 6)
```

Lambda = 0 is immediately visible as a root.

Step 2 – Factor the cubic

```
Try L = 1: 1 - 6 + 11 - 6 = 0  YES
Divide out (L-1): L^2 - 5L + 6
L^2 - 5L + 6 = (L-2)(L-3)
```

Rational root theorem: test integers 1, 2, 3, 6.

Step 3 – Full factored form

```
p(L) = L(L-1)(L-2)(L-3) = 0
```

Each factor appears exactly once.

Final Answer: lambda = 0, 1, 2, 3 — all with AM = 1 (four distinct eigenvalues)

Key Insight: Four distinct eigenvalues means the matrix is automatically diagonalizable and has exactly 4 linearly independent eigenvectors.

Question 4

Topic: Eigenvalue Computation — Trace and Determinant Shortcuts | Level: Moderate

For a 2x2 matrix A, the trace is 7 and the determinant is 12.

Find both eigenvalues without knowing the individual entries.

SOLUTION

Step 1 – Recall the 2x2 eigenvalue relations

```
For any 2x2 matrix:  lambda_1 + lambda_2 = trace(A)
                      lambda_1 * lambda_2 = det(A)
```

These follow directly from Vieta's formulas applied to the characteristic polynomial.

Step 2 – Set up and solve the system

```
lambda_1 + lambda_2 = 7
lambda_1 * lambda_2 = 12
=> lambda^2 - 7*lambda + 12 = 0
=> (lambda - 3)(lambda - 4) = 0
```

Write the characteristic polynomial directly from trace and determinant, then factor.

Final Answer: lambda_1 = 3, lambda_2 = 4

Key Insight: The characteristic polynomial of any 2x2 matrix is L^2 - tr(A)*L + det(A) = 0. Memorise this.

Section 2 Finding Eigenvectors

Once the eigenvalues are known, substitute each one into (A - lambda*I)x = 0 and solve the homogeneous system. There will always be at least one free variable because det(A - lambda*I) = 0 by construction.

Question 5

Topic: Finding Eigenvectors — 2x2, Two Distinct Values | Level: Moderate

Find the eigenvectors of A = [[3, -2], [1, 0]].

SOLUTION

Step 1 – Find eigenvalues first

```
det([[3-L, -2],[1, -L]]) = (3-L)(-L) + 2 = L^2 - 3L + 2 = 0
(L-1)(L-2) = 0  =>  L = 1 and L = 2
```

Step 2 – Eigenvector for lambda = 1

```
B = A - I = [[2, -2],[1, -1]]
Row reduce: [[1, -1],[0, 0]]
Equation: x1 - x2 = 0  =>  x1 = x2 = k
Eigenvector: k * [1, 1]
```

x2 is the free variable. Let x2 = k, then x1 = k.

Step 3 – Eigenvector for lambda = 2

```
B = A - 2I = [[1, -2],[1, -2]]
Row reduce: [[1, -2],[0, 0]]
Equation: x1 - 2*x2 = 0  =>  x1 = 2k
Eigenvector: k * [2, 1]
```

x2 is free. Let x2 = k, then x1 = 2k.

Final Answer: For lambda=1: span{[1,1]}. For lambda=2: span{[2,1]}.

Key Insight: Check: A*[1,1] = [1,1] = 1*[1,1] YES. A*[2,1] = [4,2] = 2*[2,1] YES.

Question 6

Topic: Finding Eigenvectors — Repeated Eigenvalue, Full Span | Level: Moderate

Find the eigenvectors of A = [[5, 0, 0], [0, 5, 0], [0, 0, 5]] (= 5*I).

SOLUTION

Step 1 – Find eigenvalue

```
det(A - L*I) = (5-L)^3 = 0  =>  L = 5  (AM = 3)
```

Scalar multiple of identity — every direction is an eigenvector.

Step 2 – Solve (A - 5I)x = 0

```
A - 5I = [[0,0,0],[0,0,0],[0,0,0]]
System: 0*x1 + 0*x2 + 0*x3 = 0  (always true)
All three variables x1, x2, x3 are free.
```

The zero matrix has trivial null space = all of R^3.

Step 3 – Basis for eigenspace

```
x = s*[1,0,0] + t*[0,1,0] + u*[0,0,1]
Basis: {e1, e2, e3}
```

GM = 3 = AM. This is the maximum possible.

Final Answer: Every non-zero vector in R^3 is an eigenvector. GM(5) = 3 = AM(5).

Key Insight: For a scalar matrix c*I, every non-zero vector is an eigenvector with eigenvalue c. AM = GM always.

Question 7

Topic: Finding Eigenvectors — 3x3, Three Distinct Values | Level: Moderate

A = [[0, 0, 2], [0, 2, 0], [2, 0, 0]]. Given eigenvalues are -2, 2, 2.

Find the eigenvectors for each eigenvalue.

SOLUTION

Step 1 – Verify eigenvalues via diagonal-expansion

```
det(A - L*I) = det([[-L,0,2],[0,2-L,0],[2,0,-L]])
Expanding along row 2:  (2-L)*det([[-L,2],[2,-L]])
= (2-L)*(L^2 - 4) = (2-L)(L-2)(L+2) = -(L-2)^2*(L+2)
Eigenvalues: L = 2 (AM=2), L = -2 (AM=1)
```

Step 2 – Eigenvector for lambda = -2

```
B = A - (-2)I = [[2,0,2],[0,4,0],[2,0,2]]
Row reduce: [[2,0,2],[0,4,0],[0,0,0]]  =>  [[1,0,1],[0,1,0],[0,0,0]]
x3 free = k;  x2 = 0;  x1 = -k
Eigenvector: k*[-1, 0, 1]
```

Step 3 – Eigenvectors for lambda = 2

```
B = A - 2I = [[-2,0,2],[0,0,0],[2,0,-2]]
Row reduce: [[-2,0,2],[0,0,0],[0,0,0]]  =>  [[1,0,-1],[0,0,0],[0,0,0]]
x2 free = s,  x3 free = t;  x1 = t
Eigenvectors: s*[0,1,0] + t*[1,0,1]
```

Two free variables because rank(B) = 1 and n = 3, so GM = 3-1 = 2.

Final Answer: lambda=-2: span{[-1,0,1]}. lambda=2: span{[0,1,0], [1,0,1]}.

Key Insight: GM(2) = 2 = AM(2) here because A is symmetric! Real symmetric matrices always achieve AM = GM.

Section 3 Linear Independence of Eigenvectors

Theorem: eigenvectors from distinct eigenvalues are always linearly independent. For repeated eigenvalues the independence must be verified by counting free variables.

Question 8

Topic: Linear Independence — Distinct Eigenvalues Argument | Level: Moderate

A 4x4 matrix has eigenvalues 1, 3, 3, 7 (AM shown by multiplicity).

State the minimum guaranteed number of linearly independent eigenvectors and explain why.

SOLUTION

Step 1 – Identify distinct eigenvalues

```
Distinct eigenvalues: 1, 3, 7  (three distinct values)
```

Even though 3 repeats, the set of DISTINCT values has 3 members.

Step 2 – Apply Theorem 1

```
Theorem 1: eigenvectors from DIFFERENT eigenvalues are linearly independent.
Pick one eigenvector for lambda=1, one for lambda=3, one for lambda=7.
These three are guaranteed to be linearly independent.
```

Step 3 – Consider lambda = 3 separately

```
AM(3) = 2  =>  GM(3) is either 1 or 2 (we cannot tell without more info).
If GM(3) = 2 we get a 4th linearly independent eigenvector.
If GM(3) = 1 we get only 3 total.
```

Final Answer: At least 3 linearly independent eigenvectors; at most 4.

Key Insight: 'At least' comes from distinct eigenvalue count. 'At most' comes from total matrix dimension.

Question 9

Topic: Linear Independence — Proof Sketch for Distinct Case | Level: Moderate

Suppose A*v1 = 2*v1 and A*v2 = 5*v2 with v1, v2 both non-zero.

Prove directly that v1 and v2 are linearly independent.

SOLUTION

Step 1 – Assume linear dependence (contradiction setup)

```
Suppose v1 = k*v2 for some scalar k.
```

We derive a contradiction from this assumption.

Step 2 – Multiply both sides by A

```
A*v1 = A*(k*v2) = k*(A*v2) = k*(5*v2) = 5k*v2
But also: A*v1 = 2*v1 = 2*(k*v2) = 2k*v2
```

Two expressions for A*v1 must be equal.

Step 3 – Equate and conclude

```
5k*v2 = 2k*v2
=> (5k - 2k)*v2 = 0
=> 3k*v2 = 0
Since v2 is non-zero (it's an eigenvector): k = 0
=> v1 = 0*v2 = 0  -- contradicts v1 being an eigenvector (non-zero).
```

The contradiction shows our assumption was false.

Final Answer: v1 and v2 are linearly independent. QED.

Key Insight: This argument generalises: the key step is that (lambda_1 - lambda_2)*k = 0, and since the eigenvalues differ, k must be 0.

Question 10

Topic: Linear Independence — Checking via Reduced Row Form | Level: Moderate

For matrix A = [[1,2],[2,1]], the eigenvectors are v1 = [1,1] (lambda=3) and v2 = [1,-1] (lambda=-1).

Verify linear independence directly by row reduction.

SOLUTION

Step 1 – Form matrix with eigenvectors as columns

```
P = [[1, 1],[1,-1]]
```

Columns are the eigenvectors we want to test.

Step 2 – Row reduce P

```
[[1,  1],    R2 -> R2 - R1
 [1, -1]]
[[1,  1],
 [0, -2]]
```

Two pivots found — no row of zeros.

Step 3 – Conclude

```
Two pivots in a 2x2 matrix  =>  rank = 2  =>  columns are linearly independent.
```

Final Answer: v1 and v2 are linearly independent (rank of [v1|v2] = 2).

Key Insight: This could also be verified via the determinant: det([[1,1],[1,-1]]) = -1-1 = -2 != 0.

Section 4 Arithmetic & Geometric Multiplicity

AM = power of (lambda - lambda_0) in the characteristic polynomial. GM = number of linearly independent eigenvectors = n - rank(A - lambda_0 * I). Always: 1 <= GM <= AM.

Question 11

Topic: AM & GM — Computing Both for a Repeated Eigenvalue | Level: Moderate

A = [[3, 1, 0], [0, 3, 1], [0, 0, 3]].

Find the eigenvalue, its AM, and its GM.

SOLUTION

Step 1 – Find eigenvalue and AM

```
A is upper triangular  =>  eigenvalue is 3 (all diagonal entries).
Characteristic polynomial: (3 - L)^3 = 0  =>  AM(3) = 3
```

Step 2 – Compute B = A - 3I

```
B = [[0,1,0],[0,0,1],[0,0,0]]
```

Step 3 – Row reduce B and find rank

```
B is already in row echelon form.
Pivots: position (1,2) and (2,3)  =>  rank(B) = 2
```

Step 4 – Compute GM

```
GM(3) = n - rank(B) = 3 - 2 = 1
```

Only one free variable (x1), so only one linearly independent eigenvector.

Step 5 – Find that eigenvector

```
From B*x = 0: x2 = 0 (from row 2),  x3 = 0 (from row 1 with x2=0), x1 = k
Eigenvector: k * [1, 0, 0]
```

Final Answer: Eigenvalue lambda=3: AM=3, GM=1. AM > GM (not diagonalizable over this eigenvalue).

Key Insight: This is a Jordan block — the canonical example where AM > GM. The matrix is NOT diagonalizable.

Question 12

Topic: AM & GM — Reading Bounds from Characteristic Equation | Level: Moderate

The characteristic equation of matrix A is:

(lambda - 2)^3 * (lambda - 5)^2 * (lambda + 1) = 0

State the dimension of A, the minimum and maximum number of linearly independent eigenvectors, and the constraints on each GM.

SOLUTION

Step 1 – Find matrix dimension

```
Sum of powers: 3 + 2 + 1 = 6  =>  A is 6x6
```

Step 2 – List eigenvalues and AMs

```
lambda = 2:   AM = 3
lambda = 5:   AM = 2
lambda =-1:   AM = 1
```

Step 3 – Minimum linearly independent eigenvectors

```
Three DISTINCT eigenvalues  =>  at least 3 (one from each, by Theorem 1)
```

Step 4 – Maximum linearly independent eigenvectors

```
Total matrix dimension = 6  =>  at most 6
```

Step 5 – GM constraints per eigenvalue

```
GM(2)  in {1, 2, 3}   (1 <= GM <= AM=3)
GM(5)  in {1, 2}      (1 <= GM <= AM=2)
GM(-1) = 1            (AM=1 forces GM=1)
```

Final Answer: 6x6 matrix. At least 3, at most 6 independent eigenvectors. GM constraints as above.

Key Insight: When AM=1 for an eigenvalue, GM is forced to be exactly 1 — no ambiguity.

Question 13

Topic: AM & GM — Reverse Problem: From Free Variables to AM | Level: Moderate

A is a 10x10 matrix. When solving (A - 4I)x = 0,

you find exactly 3 free variables.

(a) What is GM(4)? (b) What are the possible values of AM(4)?

(c) What is the minimum possible number of distinct eigenvalues of A?

SOLUTION

Final Answer: (a) GM(4)=3. (b) AM(4) in {3,4,...,10}. (c) Minimum = 1 distinct eigenvalue.

Key Insight: The reverse direction: given GM, AM must be >= GM. The exact AM cannot be determined without more information.

Question 14

Topic: AM & GM — Rank of (A - lambda*I) Relationship | Level: Moderate

A is a 5x5 matrix with eigenvalue lambda = 7.

You compute rank(A - 7I) = 3.

Find GM(7) and state whether AM(7) could be 2.

SOLUTION

Step 1 – Apply the rank-nullity formula

```
GM(lambda) = n - rank(A - lambda*I)
GM(7) = 5 - 3 = 2
```

GM equals the dimension of the null space of (A - lambda*I).

Step 2 – Check whether AM(7) = 2 is consistent

```
We need AM(7) >= GM(7) = 2, so AM(7) = 2 is the minimum valid value.
AM(7) = 2 is consistent (GM = AM = 2 is achievable, e.g., for symmetric matrices).
```

Step 3 – Additional constraint

```
Sum of all AMs = 5 (matrix dimension).
If AM(7) = 2, the remaining 3 eigenvalue slots belong to other eigenvalues.
```

Final Answer: GM(7) = 2. AM(7) could be 2 — it is valid (AM >= GM = 2 is satisfied).

Key Insight: rank(A - lambda*I) + GM(lambda) = n. This is a direct application of the rank-nullity theorem.

Section 5 Real Symmetric Matrices

For real symmetric matrices (A = A^T, all real entries): (1) all eigenvalues are real, (2) eigenvectors for distinct eigenvalues are orthogonal, (3) AM = GM for every eigenvalue — always n linearly independent eigenvectors.

Question 15

Topic: Real Symmetric — AM = GM Guarantee | Level: Moderate

A real symmetric 6x6 matrix has characteristic polynomial:

(lambda - 1)^2 * (lambda - 4)^3 * (lambda - 9)^1 = 0.

State exactly how many linearly independent eigenvectors exist for each eigenvalue and in total.

SOLUTION

Step 1 – State the AM = GM theorem for symmetric matrices

```
For real symmetric matrices: GM(lambda) = AM(lambda) for every eigenvalue.
```

This is the spectral theorem — no ambiguity in the 'at most' bound.

Step 2 – Assign exact GMs

```
GM(1) = AM(1) = 2  =>  exactly 2 linearly independent eigenvectors
GM(4) = AM(4) = 3  =>  exactly 3 linearly independent eigenvectors
GM(9) = AM(9) = 1  =>  exactly 1 linearly independent eigenvector
```

Step 3 – Total

```
Total = 2 + 3 + 1 = 6 = dimension of A  (confirms: always n = 6 for symmetric)
```

Final Answer: GM(1)=2, GM(4)=3, GM(9)=1. Total: exactly 6 linearly independent eigenvectors.

Key Insight: For symmetric matrices, the phrase 'at least / at most' collapses to 'exactly'. This makes symmetric matrices extremely well-behaved.

Question 16

Topic: Real Symmetric — Orthogonality of Eigenvectors | Level: Moderate

A = [[2, 1], [1, 2]] (real symmetric).

Find both eigenvalues and eigenvectors, then verify they are orthogonal.

SOLUTION

Step 1 – Find eigenvalues

```
det([[2-L,1],[1,2-L]]) = (2-L)^2 - 1 = 0
L^2 - 4L + 3 = 0  =>  (L-1)(L-3) = 0
L1 = 1,  L2 = 3
```

Step 2 – Eigenvector for lambda = 1

```
B = [[1,1],[1,1]],  row reduce: [[1,1],[0,0]]
x1 + x2 = 0  =>  x1 = -k, x2 = k
v1 = [-1, 1]  (or [1, -1])
```

Step 3 – Eigenvector for lambda = 3

```
B = [[-1,1],[1,-1]],  row reduce: [[1,-1],[0,0]]
x1 - x2 = 0  =>  x1 = x2 = k
v2 = [1, 1]
```

Step 4 – Verify orthogonality

```
v1 . v2 = (-1)(1) + (1)(1) = -1 + 1 = 0  YES
```

The dot product is zero, confirming perpendicularity — as the theorem guarantees.

Final Answer: lambda=1: v1=[-1,1]. lambda=3: v2=[1,1]. Dot product = 0 (orthogonal, confirmed).

Key Insight: This is the spectral theorem in action. A = P*diag(1,3)*P^T where P = (1/sqrt(2))*[[-1,1],[1,1]].

Question 17

Topic: Real Symmetric — Identifying from Eigenvalue Properties | Level: Moderate

A student claims: 'Matrix B has a complex eigenvalue 3 + 2i.

Therefore B cannot be real and symmetric.'

Is this claim correct? Explain fully.

SOLUTION

Step 1 – State the relevant theorem

```
Theorem: all eigenvalues of a real symmetric matrix are REAL.
```

This is one of the two fundamental properties of real symmetric matrices.

Step 2 – Apply the theorem to B

```
If B were real and symmetric, all its eigenvalues would be real.
B has eigenvalue 3 + 2i, which is complex (not real).
Therefore B cannot be real and symmetric.
```

The contrapositive of the theorem directly applies here.

Step 3 – Contrapositive logic

```
Theorem: real symmetric  =>  all eigenvalues real.
Contrapositive: NOT(all eigenvalues real)  =>  NOT(real symmetric).
B has a non-real eigenvalue  =>  B is not real symmetric.
```

Final Answer: The claim is CORRECT. A complex eigenvalue is impossible for a real symmetric matrix.

Key Insight: Contrapositive reasoning is a powerful tool. The theorem's contrapositive gives us an immediate test.

Section 6 Diagonalizability

A matrix is diagonalizable if and only if it has n linearly independent eigenvectors (equivalently: GM = AM for every eigenvalue). If diagonalizable: A = P * Lambda * P^-1 where P has eigenvectors as columns.

Question 18

Topic: Diagonalizability — Decision with Repeated Eigenvalue | Level: Moderate

Determine whether each matrix is diagonalizable:

(a) B = [[4, 0, 0], [0, 4, 1], [0, 0, 4]]

(b) C = [[4, 0, 0], [0, 4, 0], [0, 0, 4]]

SOLUTION

Final Answer: (a) B is NOT diagonalizable (GM < AM). (b) C IS diagonalizable (GM = AM = 3).

Key Insight: A single extra 1 above the diagonal (Jordan block structure) destroys diagonalizability — powerful illustration of why AM = GM matters.

Question 19

Topic: Diagonalizability — Constructing P and Lambda | Level: Moderate

A = [[1, 2], [2, 1]]. Given eigenvalues are 3 and -1 with eigenvectors [1,1] and [1,-1].

Write the diagonalization A = P * Lambda * P^-1 explicitly.

SOLUTION

Step 1 – Form P from eigenvectors (as columns)

```
P = [[1, 1], [1, -1]]
(First column = eigenvector for lambda=3, second for lambda=-1)
```

The ORDER of columns in P must match the order of eigenvalues in Lambda.

Step 2 – Form Lambda (diagonal matrix of eigenvalues)

```
Lambda = [[3, 0], [0, -1]]
```

Step 3 – Compute P^-1

```
det(P) = (1)(-1) - (1)(1) = -2
P^-1 = (1/-2) * [[-1,-1],[-1,1]] = [[1/2, 1/2],[1/2,-1/2]]
```

Use the 2x2 inverse formula: (1/det)*[[d,-b],[-c,a]].

Step 4 – Verify: P * Lambda * P^-1 = A

```
P * Lambda = [[3,1*(-1)],[3,-1]] = [[3,-1],[3,1]]
Multiply by P^-1 ... result = [[1,2],[2,1]] = A  (verified)
```

Final Answer: A = [[1,1],[1,-1]] * [[3,0],[0,-1]] * [[1/2,1/2],[1/2,-1/2]]

Key Insight: The diagonalization A = P*Lambda*P^-1 means A^k = P*Lambda^k*P^-1, making matrix powers trivial to compute.

Question 20

Topic: Diagonalizability — Using A^2 to Reason About Eigenvalues | Level: Moderate-Hard

If lambda is an eigenvalue of A with eigenvector v,

show that lambda^2 is an eigenvalue of A^2 with the same eigenvector v.

Hence find the eigenvalues of A^2 if A has eigenvalues -3, 1, 4.

SOLUTION

Step 1 – Start from the eigenvalue equation

```
A * v = lambda * v        (given: v is eigenvector of A)
```

Step 2 – Apply A again to both sides

```
A * (A*v) = A * (lambda*v)
A^2 * v   = lambda * (A*v)    (A is linear, scalar pulls out)
A^2 * v   = lambda * lambda*v
A^2 * v   = lambda^2 * v
```

This shows v is also an eigenvector of A^2, with eigenvalue lambda^2.

Step 3 – Apply to the given eigenvalues

```
A has eigenvalues: -3, 1, 4
A^2 eigenvalues:   (-3)^2=9,  1^2=1,  4^2=16
```

Step 4 – General power rule

```
By induction: if A*v = L*v, then A^k * v = L^k * v for any positive integer k.
Eigenvalues of A^k are {L^k : L is eigenvalue of A}.
```

Final Answer: Eigenvalues of A^2 are 9, 1, 16. Same eigenvectors as A.

Key Insight: This principle extends: eigenvalues of p(A) for any polynomial p are p(lambda). This is why diagonalization makes polynomial functions of matrices trivial.

Quick Reference Summary

Concept	Formula / Rule	Key Condition
Characteristic equation	det(A - L*I) = 0	Always square matrix
Eigenvalue of triangular matrix	Read from diagonal directly	Upper or lower triangular
Eigenvector computation	Solve (A - L*I)x = 0 by row reduction	Free variables give basis
Geometric Multiplicity (GM)	GM(L) = n - rank(A - L*I)	n = matrix size
Arithmetic Multiplicity (AM)	Power of (L - L0) in char. poly.	AM >= 1
AM-GM inequality	1 <= GM(L) <= AM(L)	All matrices
At least k eigenvectors	k distinct eigenvalues present	Theorem 1
At most n eigenvectors	n = matrix dimension	Always
AM = GM guaranteed	A is real symmetric	Spectral theorem
All eigenvalues real	A is real symmetric	Spectral theorem
Eigenvectors orthogonal	A is real symmetric, distinct L	Spectral theorem
Diagonalizable condition	Sum of all GMs = n	Equivalent: GM=AM for all L
Diagonalization	A = P * Lambda * P^-1	P = eigenvector matrix
Power rule	Eigenvalues of A^k = {L^k}	Same eigenvectors
Trace identity	trace(A) = sum of all eigenvalues	Counting with AM
Determinant identity	det(A) = product of all eigenvalues	Counting with AM

All 20 problems and their solutions are based exclusively on methods presented in the lecture series: characteristic equation, row reduction for eigenvectors, the AM/GM framework, and the two fundamental theorems on linear independence of eigenvectors. No Gram-Schmidt or SVD methods are used.